Fermi–Amaldi-Based Local Mixing Function

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(Dated: March 12, 2026)

I. MAIN TEXT

Consider the exact exchange energy density (closed-shell):

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = -\frac{1}{4}\int \frac{|\gamma(\mathbf{r};\mathbf{r}')|^2}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}', \tag{1}$$

where

$$\gamma(\mathbf{r},\mathbf{r}') = 2\sum_{i=1}^{N/2}\varphi_i^*(\mathbf{r})\varphi_i(\mathbf{r}') \tag{2}$$

is one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N,

$$e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = -\frac{1}{2N}\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}'. \tag{3}$$

For one-orbital systems (one-electron and two-electron singlet), two quantities are equal for all values of $\mathbf{r}$ [5–7]:

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) \quad N \leq 2 \quad \text{for all } \mathbf{r}. \tag{4}$$

For systems with more than one molecular orbital, the spatial behavior of both quantities needs to be studied closer to establish a relation similar to Eq. (4). Which is convenient to do analysing the corresponding exchange holes. The exchange hole is defined via

$$e_{\mathrm{x}}(\mathbf{r}) = \frac{1}{2}\int \frac{\rho(\mathbf{r})\rho_{\mathrm{x}}(\mathbf{r},\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}' \tag{5}$$

leading to expressions for the exact and the Fermi–Amaldi exchange holes, respectively:

$$\rho_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r},\mathbf{r}') = -\frac{1}{2}\frac{|\gamma(\mathbf{r},\mathbf{r}')|^2}{\rho(\mathbf{r})}, \tag{6}$$

$$\rho_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}') = -\frac{1}{N}\rho(\mathbf{r}'). \tag{7}$$

Both quantities are strictly non-positive, hence, we will consider their non-negative absolute values, $|\rho_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r})|$ and $|\rho_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r})|$, for convenience.

II. SCRAP

A local hybrid exchange functional is given by

$$E_{\mathrm{x}}^{\mathrm{LH}} = \int \left\{g(\mathbf{r})e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) + [1-g(\mathbf{r})]e_{\mathrm{x}}^{\mathrm{SL}}(\mathbf{r})\right\}d\mathbf{r}, \tag{8}$$

where the integrand is a mixture of two exchange density energies weighed by a local mixing function (LMF), $g(\mathbf{r})$, which is position-dependent. The $e_{\mathrm{x}}^{\mathrm{SL}}$ and $e_{\mathrm{x}}^{\mathrm{exact}}$ quantities are a semi-local and the exact exchange energy densities, respectively.

Here, we propose a new local mixing function for local hybrid functionals. This LMF is the ratio between the Fermi–Amaldi and the exact exchange energy densities:

$$g(\mathbf{r}) = \frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}}. \tag{9}$$

Both exchange densities are manifestly non-positive and equivalent for one-electron systems and two-electron singlets. Although, one can show that

$$e_{\mathrm{x}}^{\mathrm{FA}} > e_{\mathrm{x}}^{\mathrm{exact}} \quad \text{when} \quad N > 2 \tag{10}$$

by considering corresponding exchange holes. Now, compare diagonal elements ($\mathbf{r} = \mathbf{r}'$) of the exact exchange hole to the Fermi–Amaldi exchange hole:

$$\rho_{\mathrm{x}}^{\mathrm{FA}} > \rho_{\mathrm{x}}^{\mathrm{exact}}\big|_{\mathbf{r}=\mathbf{r}'} \quad \text{when} \quad N > 2. \tag{11}$$

Since, $\rho_{\mathrm{x}}^{\mathrm{exact}}$ off-diagonal elements are negative, adding them to the diagonal part makes $\rho_{\mathrm{x}}^{\mathrm{exact}}$ even more negative. Thus, for any number of electrons:

$$e_{\mathrm{x}}^{\mathrm{exact}} \leq e_{\mathrm{x}}^{\mathrm{FA}} \leq 0. \tag{12}$$

Combining these observations, we find that the proposed LMF is bound between zero and one:

$$0 \leq \frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} \leq 1. \tag{13}$$

This inequality ensures that one hundred percent of exact exchange is employed in Eq. (8) for one-electron systems, satisfying the one-electron constraint [8, 9].

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \to \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r\to\infty}\frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} = 1 \tag{14}$$

since both energy densities approach $-1/2r$ as $r \to \infty$.

Another important constraint, the high-density limit,

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